INCH-POUND

MIL-DTL-11733E
3 December 2001
SUPERSEDING
MIL-C-11733D
11 April 1974

DETAIL SPECIFICATION

CYLINDER, COMPRESSED GAS: AMMONIA, WITH VALVE (DOT 4AA480)

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of steel cylinders, with valve and cap, intended for anhydrous ammonia service.

1.2 Classification. The cylinders covered by this specification will be of the following types, as specified (see 6.2).

Type I - 100 pounds
Type II - 150 pounds

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: Defense Supply Center Richmond, ATTN: DSCR-VBD, Richmond, VA 23297-5610 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8120

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and the supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.

DEPARTMENT OF DEFENSE

MIL-DTL-2	-	Valves, Cylinder, Gas (for Compressed or Liquefied Gases).
MIL-DTL-2/8	-	Valve, cylinder, Gas: Anhydrous Ammonia (Convex-Head Cylinder) Outlet 240, Inlet .75 Inch.
MIL-C-17376/2	-	Caps and Flanges, Compressed-Gas Cylinder: Flanges, Low Pressure.

STANDARDS

FEDERAL

MIL-STD-101	-	Color Code for Pipelines and for Compressed Gas Cylinders.

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Electronic copies of military specifications and standards may be obtained from http://astimage.daps.dla.mil/quicksearch/.)

2.2.2 Other government publications. The following other government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 178.56	-	Specification 4AA480 Welded Steel Cylinders.

(Application for copies should be addressed to the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of CFR documents may be obtained from http://www.access.gpo.gov/.)

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DRAWINGS

810-1385867	-	Decalcomania for Navy Gas Cylinders.

(Copies of drawings required by suppliers in connection with a specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of documents that are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issues of documents cited in the solicitation (see 6.2).

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SSPC-SP5	-	White Metal Blast Cleaning (DoD adopted).
SSPC-SP6	-	Commercial Blast Cleaning (DoD adopted).
SSPC-SP10	-	Near-White Blast Cleaning (DoD adopted).
SSPC-Paint 25	-	Zinc Oxide, Alkyd, Linseed Oil Primer for Use Over Hand Cleaned Steel, Type I and Type II.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213. Electronic copies of SSPC documents may be obtained from http://www.sspc.org/.)

MASTER PAINTERS INSTITUTE (MPI)

MPI Reference 9	-	Enamel, Alkyd, Gloss, Low VOC Content.

(Application for copies should be addressed to the Painting & Decorating Contractors of America - National Office (PDCA), 3913 Old Lee Highway, suite 33-B, Fairfax, VA 22030-2433.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Configuration. Cylinders shall conform to requirements of 49 CFR 178.56 for a Department of Transportation (DOT) 4AA480 cylinder. The cylinders shall be within dimensions specified in table I, in accordance with figure 1, and shall be as specified herein. All dimensions in table I are in inches. The tolerance for diameter (D) is plus or minus 1/2 inch and the tolerance for height (H) is plus or minus 1 inch.

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TABLE I. Cylinder dimensions.

Type	Nominal gas capacity (pounds)	Outside dimensions		Minimum water capacity (cubic inches)	Maximum tare weight (pounds)	Valve MIL-DTL-2/8
		Diameter (D) (inches)	Height (H) (inches)			
I	100	15	38	5,140	105	V8-240-1
II	150	15	53	7,700	162	V8-240-1

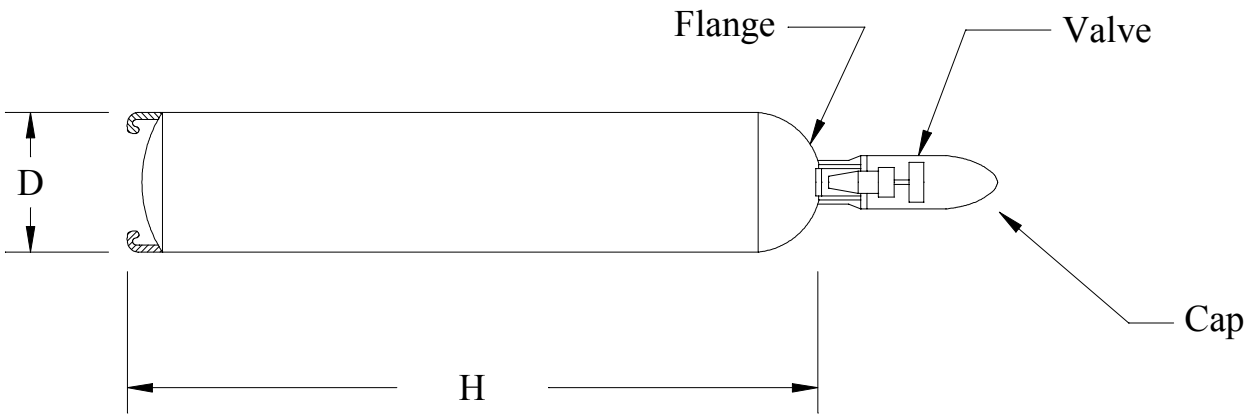


Figure 1. Cylinder configuration.

3.2 Construction. The cylinders shall be true of form and shall be stable when in an upright position. The footrings shall be attached by welding and shall have a minimum of two ventilation holes and two drainage holes equally spaced around the base of the cylinder and shall be of the same outside diameter as that of the cylinder wall.

3.3 Cylinder processing.

3.3.1 Internal surfaces. The internal surfaces of drawn cylinders, prior to necking, shall be free of mill scale, burrs, oxidation products, and all foreign matter. When required, only iron-based abrasives shall be used to remove unstable surfaces or oxidation products from the cylinder wall. Metal discoloration, rust bloom, and fixed scale shall not be considered criteria for cylinder rejection.

3.3.2 Cylinder cleaning. After necking, drilling, threading and deburring, the cylinders shall be cleared of loose particles by vacuum or by inverting the cylinder and using a compressed air sweep (see 6.3). The cylinder shall be flushed with hot fluids when necessary to remove grease, oil, and fat from processing. Chemicals that will react with anhydrous ammonia shall not be used.

3.3.3 Cylinder drying. Internal cylinder moisture, from processing or hydrostatic testing, shall be removed by inverting and steaming the surfaces until hot to the touch, followed by drying with filtered air having a dew point of less that 30 ˚F (-1 ˚C). In lieu of filtered air, drying by suction of ambient air through the cylinder is allowed. Closure shall be carried out immediately to limit atmospheric interchange.

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3.4 Closure. Cylinders shall be equipped with a steel flange in accordance with MIL-C-17376/2. Each flange shall be drilled and threaded 3/4 national gas taper (NGT) and fitted with a valve in accordance with MIL-DTL-2/8 and as specified in table I. Each cylinder shall be expressed in pounds and quarter-pounds, expressed in four-ounce increments to the nearest quarter-pound. The figures expressing pounds shall be separated from the figures expressing ounces by a dash (-).

3.5 Tare weight. The tare weight of the cylinder shall be the weight of the cylinder, flange, and valve, but excluding the valve protection cap. The tare weight shall be expressed in pounds and quarter-pounds, expressed in four-ounce increments to the nearest quarter-pound. The figures expressing pounds shall be separated from the figures expressing ounces by a dash (-).

3.6 Test date. The cylinders shall be delivered within one year of the test date stamped on the cylinder.

3.7 Cylinder identification. Unless otherwise specified herein, marking shall be in characters not less than 5/16 inch high.

3.7.1 Standard markings. Standard markings shall be plainly and permanently marked by stamping on the shoulder of each cylinder starting near the neck as follows:

a. "DOT 4AA480".
b. Serial number (see 6.6), as registered with the Office of Hazardous Materials Safety, in characters not less than 3/8 inch high for cylinders more than 6 inches in outside diameter.
c. "US GOVT", government symbol, as registered with the Office of Hazardous Materials Safety.
d. Date of test, stamped diametrically opposite the above marking.
e. The inspector's official mark.
f. Manufacturer or retester mark, near the date of test (preferably right above it).
g. The tare weight shall be stamped in a position other than in sequence with the serial number or test date and shall be preceded by the letters "TW".

3.7.2 Additional markings. In addition to the above markings, the cylinders shall bear the following markings:

a. The name of the gas, stenciled on the cylinder in accordance with MIL-STD-101.
b. When specified (see 6.2), two decalcomanias conforming to drawing 810-1385867 shall be affixed diametrically opposite one another on the sides of the cylinder, 90 degrees from the stenciled name of the gas.

3.8 Treatment and painting.

3.8.1 Cleaning. Each cylinder and cap shall be cleaned in accordance with TT-C-490. Cleaning may be accomplished by one of the following methods:

a. Chemical methods such as solvent cleaning, alkaline cleaning, acid cleaning, pickling, descaling with hydride, or paint stripping.

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b. Electrochemical chemical cleaning methods such as electrolyte alkaline, electropolishing, or electrolytic pickling.

c. Mechanical methods such as blasting, chipping, wire brushing, or grinding.

3.8.1.1 Blasting. Where blasting is appropriate, blast in accordance with SSPC-SP6, unless SSPC-SP5 or SSPC-SP10 is specified (see 6.2). Blasting shall not be used on surfaces that could be damaged by blasting or on components containing such surfaces unless protective masking has been applied.

3.8.2 Treatment. Cylinders shall be treated as soon as possible after cleaning (see 3.8.1) with zinc phosphate chemical conversion coating in accordance with MIL-T-490, type I.

3.8.3 Painting. Coatings shall be applied in an ambient temperature of not less than 50 ˚F (10 ˚C) nor more than 90 ˚F (32 ˚C). Paint shall be applied by any method that will deposit the specified film coat thickness (see 3.8.3.1 and 3.8.3.2) and comply with any applicable transfer efficiency regulations. Paint and surface shall be approximately the same temperature except when hot spray is used.

3.8.3.1 Primer. Primer conforming to SSPC-Paint 25 shall be applied to each dry, clean cylinder as soon as practicable after cleaning and treatment (see 3.8.1 and 3.8.2). Dry film thickness of primer shall be 0.7 to 1.0 one-thousandth of an inch (mil).

3.8.3.2 Finishing coat. A finishing coat conforming to MPI Reference 9 shall be applied to each cylinder, color conforming to the cylinder color code requirement of MIL-STD-101. The dry film thickness of finishing coat shall be 1.0 to 1.5 mil.

3.9 Workmanship. Cylinder, valves, flanges, and caps shall be clean and free from grit and loose scale. Edges shall be rounded or chamfered. Cylinders shall be clean and free of other surface defects detrimental to the intended use.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as conformance inspection.

4.2 Examination. Each cylinder shall be examined for defects as listed in table II. The presence of one or more defects shall be cause for rejection.

4.2.1 Tests. All tests shall be completed within one year from the date of acceptance by the government.

4.2.1.1 Lot. A lot shall consist of not more than 200 cylinders produced from the same heat of steel, with the same treatment cycle, offered for delivery at the same time.

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TABLE II. Processing defects.

Defect	Requirement
Cylinder not in accordance with 49 CFR 178.56 specification for a DOT 4AA480 cylinder	3.1
Dimensions not as specified	3.1
Construction not as specified	3.2
Cleaning and drying not as specified	3.3.2 and 3.3.3
Closure not as specified	3.4
Tare weight not as specified	3.5
Test date more than one year prior to the date of acceptance by the government	3.6
Markings not as specified	3.7.1 and 3.7.2
Treatment and painting not as specified	3.8
Workmanship not as specified	3.9

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel shall contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activity within the military department or defense agency, or within the military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cylinders covered by this specification are intended for use in shipment and storage of anhydrous ammonia.

6.1.1 Military-unique rationale. These cylinders may be subjected to harsh environments, including temperature extremes and salt water corrosion. For this reason, they require surface preparation and painting beyond that normally provided for commercial cylinders. Additionally, the valves, which are included as an integral part of the cylinders, are constructed to standards that exceed those manufactured for commercial customers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Cylinder classification (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. Decalcomanias, when required (see 3.7.2).

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e. Blasting specification, if appropriate (see 3.8.1.1).
f. Packaging requirements (see 5.1).

6.3 Loose particulate matter in stored cylinders. Cylinders that have been cleaned, dried, valved, and subjected to storage and handling within delivery time limits will be acceptable for ammonia service if accumulated loose ferro-related particulate matter resulting from handling does not exceed 1 gram per 1500 cubic inches of cylinder capacity.

6.4 Part or identification number (PIN). The PIN to be used for cylinders acquired to this specification is created as follows:

M11733 – 100 Example of PIN:
- 100 — Container capacity in pounds (see table I)
- M11733 — CID number

M11733-100 indicates: 100-pound container capacity.

6.5 Cross-reference. The differences in cataloging data for chlorine cylinders between this revision and MIL-C-11733D are listed by capacity as follows:

TABLE III. Cross-reference table.

Nominal gas capacity (pounds)	MIL-C-11733D	MIL-DTL-11733E
100	C-11733-1	M11733-100
150	C-11733-2	M11733-150

6.6 Serial numbering. Cylinder serial numbers will be prefixed by a two- or three-digit symbol designating the procuring agency and will have a two-letter suffix designating the supplier furnished by the contracting officer. The supplier will assign serial numbers to the individual cylinders. These numbers may be consecutive with the supplier's regular production numbers or of a series established specifically for a customer's cylinders. However, all cylinders on a given contract will be numbered consecutively and controls will be exercised to preclude duplication with future deliveries to the government.

6.7 Subject term (key word) listing.

anhydrous
cap
container
decalcomania
flange

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Air Force - 68

Preparing activity:
DLA - GS3

(Project 8120-1051)

Reviewer:
Navy - MC

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-11733E	**2. DOCUMENT DATE (YYYYMMDD)** 20011203

3. DOCUMENT TITLE

CYLINDER, COMPRESSED GAS: AMMONIA, WITH VALVE (DOT 4AA480)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Supply Center Richmond	b. TELEPHONE *Include Area Code)* Commercial: (804) 279-5019 DSN: 695-5019
c. ADDRESS *(Include Zip Code)* ATTN: DSCR-VBD (C. Hammond) 8000 Jefferson Davis Highway Richmond, VA 23297-5610	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE STANDARDIZATION PROGRAM OFFICE (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Commercial: (703) 767-6888 DSN: 427-6888

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